

Biology 005B: Introduction to Organismal Biology

Summer 2026 🌸 Spieth Hall 2200 🌸 TR 4:00–5:20pm

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[Canvas Page](#)

Week	Date	Unit	Topic
1	T (06/23)	Life as systems	1: What does it mean to be alive?
	R (06/25)		2: Well, it's complex
2	T (06/30)		3: Constraints
	R (07/02)		*4: Trade-offs
3	T (07/07)	Interdependent systems	Unit 1 Quiz: Lectures 1–4
	R (07/09)		5: The eukaryotes
4	T (07/14)		6: Multicellularity
	R (07/16)		7: From water to land
5	T (07/21)		*8: Asexual & sexual reproduction
	R (07/23)		Unit 2 Quiz: Lectures 5–8
6	T (07/28)	Self-regulating systems	9: Energy in gradients
	R (07/30)		10: Circulation & gas exchange
7	T (08/04)		11: Osmoregulation
	R (08/06)		*12: Behavior
8	T (08/11)	Systems among systems	Unit 3 Quiz: Lectures 9–12
	R (08/13)		13: Development & ontogeny
9	T (08/18)		14: Coevolution, cooperation, & coordination
	R (08/20)		15: The supraorganism
10	T (08/25)		*16: Life through time
	R (08/27)		Unit 4 Quiz: Lectures 13–16
	Sa (08/29)		Take-home final or final project

*Denotes Activity Days.

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Overview

Course description & learning outcomes

Go outside and touch grass. Feel how the blade of grass is flexible and slender, and how its tangled roots trap the damp soil between each fiber? Why does grass look like that? How does the way grass looks affect its survival and reproduction within its environment? *Biology 005B: Introduction to Organismal Biology* expands on the proximate processes you learned in BIOL 005A and foreshadows the ultimate processes you will learn in BIOL 005C to tackle the mystery of why organisms on Earth look and function so differently. Specifically, by the end of this course, you should be able to:

- ✿ Use complex systems theory as a conceptual framework to define "organism."
- ✿ Describe the concept of "emergent features" and their relevance to understanding living systems.
- ✿ Appreciate the organism as the intersection of physics, chemistry, evolution, and ecology at several levels of hierarchical organization.
- ✿ Hypothesize the ways in which an organism's form is crafted by proximate and ultimate causes, and relate this form to its function.
- ✿ Discuss the importance of cooperation in generating Earth's biodiversity.
- ✿ Evaluate the role of human social context in establishing and maintaining biological paradigms, specifically those in organismal biology.
- ✿ Make and support a scientific argument with the appropriate conceptual frameworks, examples, and other scientific justifications.

All materials will be provided to you, so no need to purchase anything! Just come to class with your brain, your heart, and sometimes a pencil. There won't be many times in your life where you get to experience such a breadth of content in a small-class setting, so make the most of this chance! Immerse yourself and make some friends.

Community guidelines

Education is liberation. Having knowledge and, more importantly, being able to critically apply this knowledge enables you to shape your world. But for education to be liberating, a learning space must be built upon love, community care, and dialogue. To ensure our learning space encompasses all three of these interconnected values, we have the following community guidelines in place:

- ✿ **Go out of your way for others.** Maintaining a vibrant community requires everyone to give a little, to meet each other halfway (a village cannot exist without villagers!). Hold the door, answer your classmate's question, check in with each other, etc.!
- ✿ **Give and assume good will.** It goes without saying that we should be nice to each other. However, people are scraggly, like scones. Sometimes words might come out wrong, or a person is still learning. When you assume good will, you give that person the ability to grow without shame, which leads me to my next guideline...
- ✿ **Fear not embarrassment and cringe.** Learning is a tough process that necessitates being exposed to new things. There will be times when you struggle—I know I sure experience those! Being confused is not embarrassing. Showing you care is not cringe. Ask that question! Geek out!

This list is ever-growing and tailored to student needs. If you have a community guideline you would like to add, please let me know!

Graded items & course grade cutoffs

A breakdown of the possible points you can receive for each assignment can be found in the table below. More details as to what each assignment entails can be found in their respective section of the syllabus.

	Assignment	Points
	Check-in assignments (5pts each x3)	15
	Poll Everywhere	35
	Activity Days (10pts each x3)	30
	Quizzes (80pts each x3)	240
	Final exam/project	80
	Lab total	100
Course total		500

A table of total point ranges and their corresponding letter grades can be found below.

Grade	Points
A+	485-500
A	470-484.99
A-	450-469.99
B+	435-449.99
B	420-434.99
B-	400-419.99
C+	385-399.99
C	370-384.99
C-	350-369.99
D+	335-349.99
D	320-334.99
D-	300-319.99
F	299.99 or below

Check-in assignments

I want to give you the best possible learning experience in BIOL 005B ☺. In order to do so, there are three check-in assignments. Each assignment is worth five points.

- 🌀 **Welcome survey.** I want to get to know you and tailor the class to your needs! The welcome survey will ask about your demographics, experience in biology, and expectations for the class. You can answer as many or as few questions as you like.

- ✿ **Syllabus quiz.** We'll work best together when you also have a clear sense of my expectations for you. Everything you need to know is in this syllabus! The syllabus quiz will recap this document's most important takeaways. You can take this quiz as many times as you need to get the full points.
- ✿ **Farewell survey.** I am always trying to grow as an educator! For me to do that, you need to tell me how I did, and you can do so on the farewell survey. You can answer as many or as few questions as you like. This survey is separate from the formal evaluations conducted by the university (which I would still like you to fill out).

Poll Everywhere

People learn the best (1) when they have a lot of opportunities to practice, and (2) when the environment is low-stakes enough for them to make mistakes without repercussions. Every lecture (except **Activity Days**), I will show 1-3 practice questions that you can answer on the Poll Everywhere app or **website**. These questions provide you with the opportunity to test your knowledge of our learning objectives. They also give you an idea of the types of questions I would ask on an exam. Each practice question you answer correctly will earn you 3 points, while any other response will earn you 1.5 points. Note that you can only earn a total of 35 PollEverywhere points for the entire quarter, so you can make many mistakes (and even miss a few lectures) without negatively affecting your grade!

Activity Days

In your future profession, you will inevitably have to piece together a series of seemingly miscellaneous facts under a conceptual framework in order to solve a problem. A patient might present mysterious symptoms, and you have to figure out the cause and cure. Or a dataset might present weird patterns, and you have to figure out what it means for the real world. Actually, forget about our fixation on labor entirely: when you're out and about, you might see things that simply have you asking, "Why?" Our Activity Days are meant to get you exposed to the questions we as biologists face. You will work with your peers to develop the sort of thought processes a biologist would use to come to a solution.

Activity Days always take place one lecture before a quiz. Activities are graded on attendance and effort, not necessarily correctness. Your lowest Activity Day grade will be dropped at the end of the quarter. As such, you may miss one Activity Day without negatively affecting your grade. Use this opportunity wisely! Because Activity Days are collaborative, it is very difficult to provide makeups unless there are extenuating circumstances. However, please do communicate with me if emergencies cause you to miss more than one Activity Day.

Quizzes

The traditional two-midterms-and-a-final structure of courses is not conducive to learning because it does not give students feedback frequently enough—by the time you realize you don't really understand something, half the quarter has already passed! As such, this class has small quizzes instead of big exams.

Furthermore, quizzes will be completed in two stages: in the first stage, you complete the quiz silently and independently; in the second stage, you complete the same quiz again in collaboration with a group of your choosing. Your final quiz score is a composite of your individual and group scores. You can calculate your final quiz score using this formula: final

score = $0.8 * (\text{individual score}) + 0.2 * (\text{group score})$. However, if your individual score is higher than your group score, your final quiz score will only consist of your individual score. I will reach out privately to discuss logistics with students who have testing accommodations.

Quizzes consist of 5 true/false questions (1 point each), 28 multiple-choice questions (2.5 points each), and one free response question (5 points), for a total of 80 points per quiz. Quizzes are not cumulative; however, Quizzes 2-4 will have three bonus questions worth one extra credit point each. These bonus questions will come from previous exams.

Your lowest quiz score will be replaced by the score you earn on your final, given your final exam/project score is higher than your lowest quiz score. Thus, you may miss one quiz without negatively affecting your grade. Again, use this opportunity wisely! Because quizzes are collaborative, it is very difficult to provide makeups unless there are extenuating circumstances. However, please do communicate with me if emergencies cause you to miss more than one quiz.

You can always review your quizzes on Gradescope, accessible through the course [Canvas Page](#).

The final

Finals are meant to exhibit your mastery in course content. Different people will prefer different ways to demonstrate their knowledge. As a result, **you may choose between two options for a final: a traditional final exam or a creative final project.**

If you choose to take the final exam, you will do so online, over Canvas. The final exam will consist of 5 true/false questions (1 point each) and 30 (2.5 points each) multiple choice questions randomly selected from previous quizzes. Each person will have a different version of the final exam. The final exam is untimed. You may use your own notes and any materials posted on the course Canvas page, but you may not collaborate with your peers, search the internet, or use AI. **The final exam will open at 3:30pm on Friday, August 28th, and close at 5:30pm on Saturday, August 29th.**

If you choose to complete a final project, please refer to the Final Project Guidelines document posted on Canvas for more specific instruction. **The final project will be due at 5:30pm on Saturday, August 29th.**

If your final exam/project score is lower than your lowest quiz score, the final will not count towards your overall course grade.

Lab

I have no control over your lab grade! Lab activities are completely separate from lecture, so your lab grade is determined by your TA and Academic Coordinator. All I do is transfer your lab grade from their Canvas page to mine in order to calculate your final grade. Please reach out to your TA or Academic Coordinator regarding questions about lab.

Course Policies

Instructor contact & office hours

You can contact me on [Canvas](#) or via your UCR email address. If you email me, please include the class name (BIOL 005B) in the subject line so I know who you are. You can expect a response within one or two business days (as I am also a person with other duties!).

You do not need to inform me if you need to miss lecture. If you need to miss an Activity Day or a quiz, please refer to the [Activity Days](#) or [Quizzes](#) sections of the syllabus.

I hold office hours on Tuesdays and Thursdays from 2:30–3:30pm. You can find my office in Spieth Hall 1360 and on [Zoom](#). Spieth Hall can be a little tricky to navigate, so please use [these directions on Google Slides](#) to find my office. Feel free to come chat about the course or anything else in life!

Grade posting & late policy

I will try to return grades within one week of the submission date. After that, you have one week to let me know of any issues related to lecture grades. Grades will be available on the course Canvas page and on Gradescope. I do statistical analysis on each exam and adjust exam grades if necessary, but I will never change the overall grade cutoffs. Your grade on Canvas is your grade!

Late assignments will receive a 15% deduction for each day these assignments are submitted past their due date (does not apply to Farewell Survey or final exam/project). However, if you have an extenuating circumstance, please do work with me to find a compromise!

Artificial "intelligence"

I put "intelligence" in quotes because I think there are a lot of misconceptions about how AI—specifically, generative AI—actually works. AI cannot think. All AI can do is algorithmically piece together fragments of stuff based on the probability that one thing comes after another. For example, the word "are" has a much higher probability of coming after the word "you" than the word "not" would, so AI is likely to spit out the string "you are" as opposed to "you not." AI calculates these probabilities based on sentences written by actual humans, humans who did not consent to have their work be used and stolen by these algorithms. AI does the same with drawings, music, stories—anything good at all that is uniquely and beautifully human.

As such, AI is not capable of creating novel ideas. If the purpose of an education is to develop the skills that enable you to solve problems nobody has ever encountered before, using AI is not going to help you get an education, nor will it prepare you for your future career and humanity. I am empathetic that you—as a student, a friend, an employee, a caretaker, a *person*—often have a lot going on. Sometimes school cannot be your first priority. But I implore you to accept the consequences of the trade-offs that come with being alive. Don't cheat your way out of a learning experience. Work with me to come to a solution for whatever circumstance is driving you to use AI.

Sometimes students will also mention that they use AI to develop study materials. I appreciate those students for trying to enhance their learning experience. But to those students I also say: there is nothing you can learn from AI that you can't learn from another human

being. Form a study group! Lean on your community. At the end of the day, all we have is each other.

If it weren't clear at this point, I vehemently hate generative AI. I feel the need to make my stance explicit because AI is a sociopolitical issue. Neutrality towards something that not only degenerates human culture, but also exposes us to a number of security issues, kills our environment, exacerbates wealth inequality, etc., is dangerous. You will learn in this course that nothing exists in a vacuum. The issue of AI is no exception.

University Policies

COVID-19

If you have been exposed to COVID-19 you must fill out the [Daily Wellness Check Survey](#). All students in this course, as a condition of physical presence in the classroom (including for exams or tests in the classroom or other location on the UCR campus), must be compliant with the [UC SARS-CoV-2 \(COVID-19\) Vaccination Program](#) at all times, as well as all other UCR COVID-related policies. Compliance is easily achieved by providing proof of Full Vaccination or submitting a request for Exception or Deferral as required by the Program. Students in violation of the Program or related directives of the instructor will be subject to disciplinary or other remedial action. The Policy can be found at: <https://policy.ucop.edu/doc/5000695/VaccinationProgramsPolicy>. More information on current COVID policy can be found at <https://campusreturn.ucr.edu/>.

Library Online

Access digital materials and other resources at the [UCR Library](#).

ITS Help and Student Technology Support

ITS Student Technology Services supports 9 student computer labs, including 7 public labs and 2 nonpublic labs, with approximately 293 public lab hours per week (with reduced hours during breaks and summer) available for academic use by all UCR students.

Whether you need a machine to work on your paper, need to use specialized software required by your class, assistance with your R'Mail or iLearn accounts, getting on the network, or any other services used by UCR students, contact bearhelp@ucr.edu.

Academic Integrity and Misconduct

Honesty and integrity are fundamental values that guide and inform us as individuals and as a community at UCR. In this class, we expect that students will take responsibility for their own learning and that they will familiarize themselves with the Principles of Academic Integrity as formulated by the Academic Senate of the University of California. Academic misconduct is any act that does or could improperly distort student grades or other student academic records. This includes cheating, fabrication, plagiarism, facilitating academic dishonesty, unauthorized collaboration, interference, and sabotage. Students found guilty of any academic misconduct will receive a reduced final course grade. For additional information please, refer to the [Academic Senate Guidelines](#).

In addition to the course conduct policies outlined by your instructor in the Course Syllabus in the online classroom, please review the Student Handbook. It is the responsibility of each student to be familiar with the definitions, policies, and procedures concerning academic misconduct. Please revisit our [Academic Integrity Policies and Procedures](#) for more information. This site also defines misconduct, provides examples of prohibited conduct, and explains the sanctions available for those found guilty of misconduct.

Plagiarism

Plagiarism is the most common form of academic misconduct at UCR. It is the appropriation of another person's ideas, processes, results, or words without giving appropriate credit.

This includes the copying of language, structure, or ideas of another and attributing (explicitly or implicitly) the work to one's own efforts. Plagiarism means using another's work without giving credit. In our course, plagiarism also includes the use of AI-assisted text without stating its explicit use. For more information about plagiarism, see [Academic Integrity Policies and Procedures](#).

Student Accommodations

UC Riverside is committed to providing equal access to learning opportunities to students with documented disabilities. To ensure access to this class, and your program, please contact the Student Disability Resource Center (SDRC) to engage in a confidential conversation about the process for requesting accommodations in the classroom. More information can be found on sdrc.ucr.edu. If you are a student registered with the SDRC, please ensure you request your quarterly accommodations through rability.ucr.edu.

Reasonable Accommodation for Disabilities

UCR is committed to providing reasonable accommodation for all students with disabilities. Students with disabilities who require accommodations in this course should contact their professor as early in the semester as possible. Students with disabilities must be registered with the Student Disability Resource Center prior to receiving accommodations in this course.

If you have a disability and you would like to make a request for reasonable accommodation, please see the Graduate Student Handbook or get in touch with the [Student Disability Resource Center](#).

Adjustments for Pregnancy/Childbirth Related Issues

Should you need modifications or adjustments to your course requirements because of documented pregnancy-related or childbirth-related issues, please contact me as soon as possible to discuss your options. Generally, modifications will be made where medically necessary and similar in scope to accommodations based on temporary disability. Learn more about the rights of pregnant and parenting students by consulting the [Office of Diversity, Equity, and Inclusion](#).

Title IX Resources

For any concerns regarding gender-based discrimination, sexual harassment, sexual misconduct, stalking, or intimate partner violence, the University offers a variety of resources, including advocates on-call 24/7, counseling services, mutual no contact orders, scheduling adjustments, and disciplinary sanctions against the perpetrator. Please see the [Title IX website](#) for more information, or call (951) 827-7070. You can also [file a report](#).

Student Needs

[Student Health Services](#), [Counseling & Psychological Services \(CAPS\)](#), [Residential Life](#), [Dining](#), and [R'Pantry](#) are available to support students.

Religious Holidays

It is the policy of the University to make a reasonable attempt to accommodate religious observances. It is the responsibility of the student to make alternate arrangements with the instructor at least one week prior to the actual date of the religious holiday, or during the first

week of classes if the holiday occurs during an exam.

Copyright Policy

It is illegal to download, upload, reproduce, or distribute any copyrighted material, in any form and in any fashion, without permission from the copyright holder or his/her authorized agent. UCR expects all members of its community to comply fully with federal copyright laws.

Registration and Withdrawal

If you choose to withdraw from this course, you must complete the appropriate University form and turn the form in before the deadline. Deadlines are shown in the [Academic Calendar](#), which is available from the Office of the Registrar.

Acknowledgments

Though I have modified this course to better fit a small-class setting and to reflect my personal teaching philosophy, much of the syllabus and structure of this course were adapted from Dr. Lucy E Delaney's BIOL 005B course. The [University Policies](#) section was taken directly from Lucy's syllabus with permission. Some course content was also adapted from Dr. Natalie C. Holt's graduate-level organismal biology course. Thanks Lucy and Natalie for being such exemplary role models! ♡